

HACKATHON 2026

Trade Surveillance & Alert Triage Engine

Monitors trading activity in real time, detects suspicious patterns using AI, and auto-triages alerts to cut false positives for compliance teams.

Duration	Friday 6 PM – Sunday 4 PM (June 4 – 6, 2026 approx. 46 hrs)
Team Size	3 – 4 members per team
LLM Stack	Anthropic Claude API (one key provided per team)
Domain	Financial Services / Capital Markets – Trade Surveillance & Compliance

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OVERVIEW

Financial firms generate thousands of trade surveillance alerts daily — the vast majority are false positives that drain compliance analyst time and delay genuine escalations. Teams will build an AI-powered surveillance assistant that ingests simulated trade and order data, detects potentially suspicious patterns, uses Claude to reason over flagged events and produce a human-readable triage verdict, and triggers automated downstream workflows to route genuine alerts to the right teams.

CORE DELIVERABLES

Task 1 – Trade Data Ingestion	Build a pipeline to ingest order and trade event data. The system must support loading a dataset of transactions and replaying them for demonstration purposes.
Task 2 – Suspicious Pattern Detection	Detect anomalous or potentially manipulative trading behaviour from the ingested data. The system must identify and flag at least two distinct types of suspicious patterns with a severity classification.
Task 3 – AI-Driven Alert Triage	For each flagged event, use the Anthropic Claude API to analyse the alert, assess whether it represents genuine misconduct or a false positive, and produce a clear human-readable verdict with a confidence score and supporting rationale.
Task 4 – Automated Escalation Workflows	On completion of triage, automatically trigger at least two downstream actions — such as creating a compliance case, sending a notification, or updating a watchlist — based on the severity and confidence of the alert.

EXAMPLE SYSTEM OUTPUT

Alert ID: TRD-2026-0042 **Timestamp:** 2026-05-09 09:47:33 UTC **Trader:** T-4821 **Instrument:** HDFC Bank (NSE: HDFCBANK) **Severity:** HIGH **Detected Pattern:** Layering / Order Book Manipulation Trader placed 14 large buy orders (avg 50,000 shares) between 09:44-09:47 12 of 14 orders cancelled within 800 ms of placement 2 sell orders executed at elevated price during cancellation window **AI Triage (Claude):** The order pattern is consistent with layering: large visible orders inflate perceived demand, inducing price movement, before cancellation enables a profitable sell-side fill. Cancellation ratio: 85.7% | Time-to-cancel median: 620 ms | Anomaly vs 30-day baseline: +4.2σ **Triage Verdict:** ESCALATE | Detection Confidence: 91% | False Positive Probability: 9% **Automated Actions Triggered:** 1. Compliance case created – Jira: COMP-8812 assigned to Surveillance Desk L2 2. Alert digest sent to #compliance-alerts Slack channel 3. Trader T-4821 flagged in watchlist for 72-hr enhanced monitoring

EVALUATION CRITERIA		
Criteria	Description	Weight
AI Triage Quality	Accuracy of Claude-generated triage verdicts, confidence scores, and <u>plain-English reasoning</u>	25%
Pattern Detection	Coverage and precision of suspicious pattern detection; false positive suppression rate	20%
Automation & Workflow	Effectiveness of post-triage escalation actions and end-to-end pipeline reliability	20%
Working Demo	Live replay of a complete scenario: ingest → detect → triage → escalate	20%
API Efficiency	Minimal and purposeful Claude API calls; prompt quality and token cost awareness	10%
Docs / README	Clarity of markdown documentation, architecture diagram, and setup instructions	5%

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